

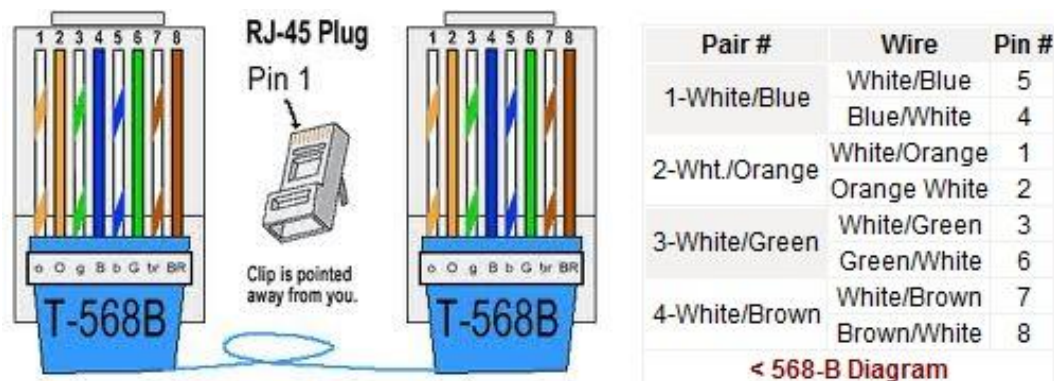
Lab 1

Cable Making

Objective:

The purpose of this lab is for the student to create a one-meter CAT5E straight thru network cable. Students will utilize various tools to create a proper network cable. Students will also utilize a network cable tester to verify that their cable is functioning properly.

Diagram:



Components:

- 1 – Spool of Cat 5E Wire
- 2 – RJ-45 Plugs

Questions:

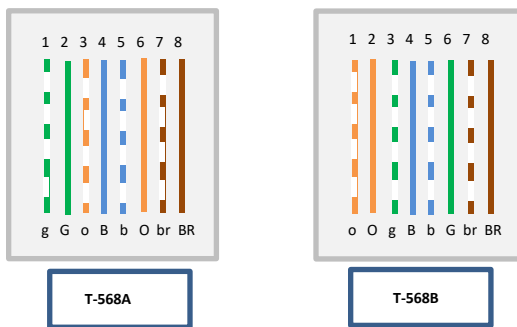
1. What tools did you utilize to complete this lab?

- a. _____ Cable stripper_____
- b. _____ Crimper_____
- c. _____ RJ45 plugs_____
- d. _____ CAT 5E cable_____
- e. _____ Cable tester_____

2. What are the steps you used to complete the cable?

- a. _____ Cut the cable length_____
- b. _____ Strip off the plastic_____
- c. _____ Untwist the wires_____
- d. _____ Arrange the wires in the proper order__
- e. _____ Straighten out the wires to get kinks out__
- f. _____ Put the wires into the RJ45 plug_____
- g. _____ Use the crimper to crimp the ends_____
- h. _____ Test the cable_____

3. Draw both ends of a CAT5B crossover cable. Label the color codes and pin numbers on both ends. You can use MS Paint or just find an image in Google images.



4. Get pricing for 100 feet of CAT5E Network cable and 20 RJ-45 plugs. I like to use www.cdw.com. Their site is easy to navigate.

- Price of wire ____ \$191.99 for 1000ft, so \$19.20 for 100 ft ____
- Price of RJ – 45 plugs ____ \$21.99 for 50, so \$0.44 each ____
- Cost per cable if 10 (10ft) cables are made __ \$2.80 (\$1.92 + .88) __
- Cost of 1 (10ft) premade cable ____ \$6.99 ____
- Which is the better value? ____ Making your own is technically less expensive, but labor intensive, and the bulk products available are more than you need to purchase, so you end up spending more. I would personally say that the premade cable is worth the extra cost. ____

5. What did you learn by completing this lab?

__ I learned how to make a functional Cat5e straight through cable, the difference between it and a crossover cable, and how to use the tools necessary to create the cable. ____